

Advanced Machine Learning – Complete Viva Cheat Sheet

Clustering • Anomaly Detection • Dimensionality Reduction • Recommenders • Time Series (ARIMA)

CLUSTERING

Clustering: Grouping similar data points without labels (unsupervised learning).

Hard vs Soft Clustering: Hard assigns one cluster per point (K-Means); Soft assigns probabilities (GMM).

Lloyd's Algorithm: Assign points → recompute centroids → repeat until convergence.

WCSS: Sum of squared distances within clusters; lower WCSS means tighter clusters.

Elbow Method: Optimal k at sharp change in WCSS slope.

Silhouette Score: Measures cluster fit; range -1 to +1.

K-Means Drawbacks: Needs k, sensitive to initialization, fails for non-globular clusters, sensitive to outliers.

K-Means++: Smarter centroid initialization to improve convergence.

HIERARCHICAL CLUSTERING

Builds a dendrogram by iteratively merging closest clusters.

Linkages: Single (min), Complete (max), Centroid, Ward (minimizes WCSS increase).

Pros: No k required, good visualization. Cons: Expensive, cannot incrementally add points.

DBSCAN

Density-based clustering using eps and minPts.

Core: $\geq$ minPts; Border: near core; Noise: isolated.

Pros: No k, arbitrary shapes, handles noise. Cons: Poor in high dimensions, parameter sensitive.

GAUSSIAN MIXTURE MODEL (GMM)

Probabilistic soft clustering using Gaussian distributions.

Uses mean and covariance; covariance defines shape and orientation of clusters.

Better than K-Means for elliptical clusters.

ANOMALY DETECTION

Anomaly: Rare abnormal point; Novelty: Unseen new pattern.

RANSAC: Robust parameter estimation using random subsets to ignore outliers.

Elliptic Envelope: Assumes unimodal Gaussian; far points are outliers.

Isolation Forest: Random splits isolate outliers faster (shallower depth).

LOF: Compares local density with neighbors; LOF $\gg$ 1 indicates outlier.

DIMENSIONALITY REDUCTION

PCA: Linear dimensionality reduction maximizing variance.

Feature Selection vs Extraction: Selection chooses subset; Extraction creates new features.

t-SNE: Non-linear visualization preserving local structure; not for modeling.

Crowding Problem: High-dim points collapse; t-SNE uses probabilities to fix.

RECOMMENDER SYSTEMS

Recommender System: Predicts items user may like to improve engagement.

Types: Content-based, Collaborative Filtering, Popularity-based, Hybrid.

Market Basket Analysis: Finds frequently bought-together items.

Apriori: Frequency-based association rule mining.

Support: Frequency of itemset; Confidence: $P(Y|X)$; Lift: $\text{Confidence}/\text{Support}(Y)$.

Confidence vs Lift: Lift corrects popularity bias.

CONTENT-BASED FILTERING

Recommends items similar to user history using item features.

Pros: No cold start, no sparsity. Cons: Low diversity, needs feature engineering.

COLLABORATIVE FILTERING

Uses behavior of similar users/items.

Item-based: 'Users who liked this also liked...'

User-based: 'Users similar to you liked...'

Similarity Metrics: Hamming, Euclidean, Cosine.

Cold Start: New users/items lack history.

MATRIX FACTORIZATION

Decomposes user-item matrix into latent factors.

Latent features represent hidden preferences.

Prediction via dot product of user and item vectors.

Pros: Handles sparsity, enables serendipity. Cons: Cold start, expensive.

RECOMMENDER EVALUATION

Precision@K: $\text{Relevant recommended} / \text{total recommended}$.

Recall@K: $\text{Relevant recommended} / \text{total relevant}$.

Serendipity: Ability to recommend unexpected but relevant items.

TIME SERIES BASICS

Time Series: Data indexed by time.

Trend: Long-term movement; Seasonality: repeating pattern; Residual: noise.

Additive: $y = \text{trend} + \text{seasonality} + \text{error}$.

Multiplicative: $y = \text{trend} \times \text{seasonality} \times \text{error}$.

FORECASTING METHODS

Mean Forecast: Average of past values.

Naïve: Last observed value.

Seasonal Naïve: Last season's value.

Drift: Linear extrapolation.

Moving Average: Average of last k values.

EXPONENTIAL SMOOTHING

SES: Level only.

DES (Holt): Level + trend.

TES (Holt-Winters): Level + trend + seasonality.

STATIONARITY & ARIMA

Stationarity: Mean and variance constant over time.

ADF Test: $p\text{-value} < 0.05$ indicates stationarity.

Differencing: $y_t - y_{t-1}$ removes trend.

ACF: Identifies MA order; PACF: Identifies AR order.

AR(p): Depends on past values; MA(q): Depends on past errors.

ARMA: AR + MA for stationary data.

ARIMA(p,d,q): Handles non-stationarity via differencing.

SARIMA(p,d,q)(P,D,Q,s): Adds seasonal components.

Residuals should be white noise; use AIC/BIC for model selection.